

Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Cómputo en la nube

4.10 Análisis de tráfico de red con Wireshark para la prevención de ataques

Docente: Ing. Miguel Salazar del Bosque

Equipo:

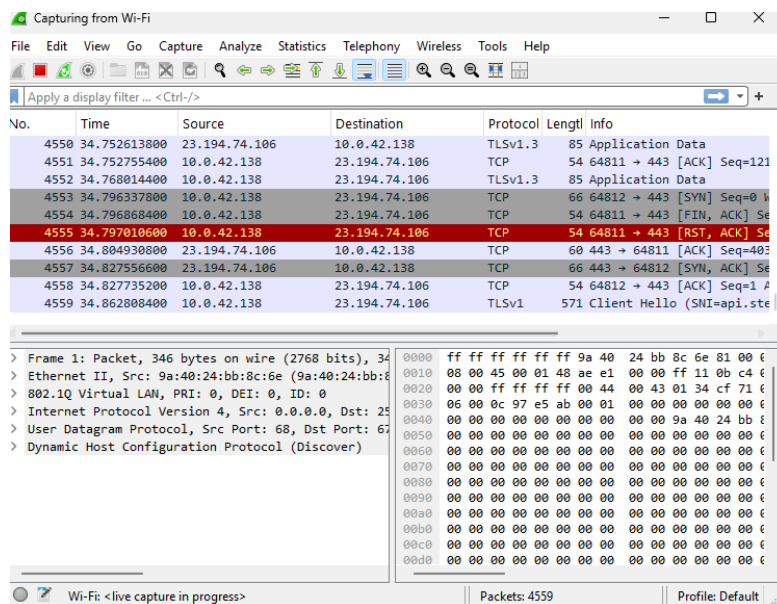
Mauro Rodrigo Ruiz Alvarez 22050727

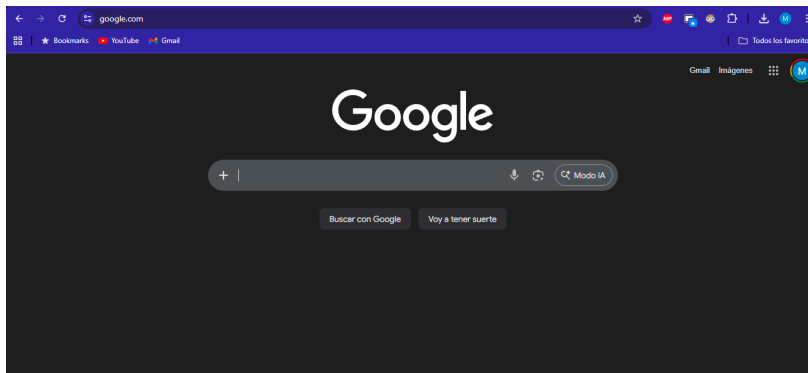
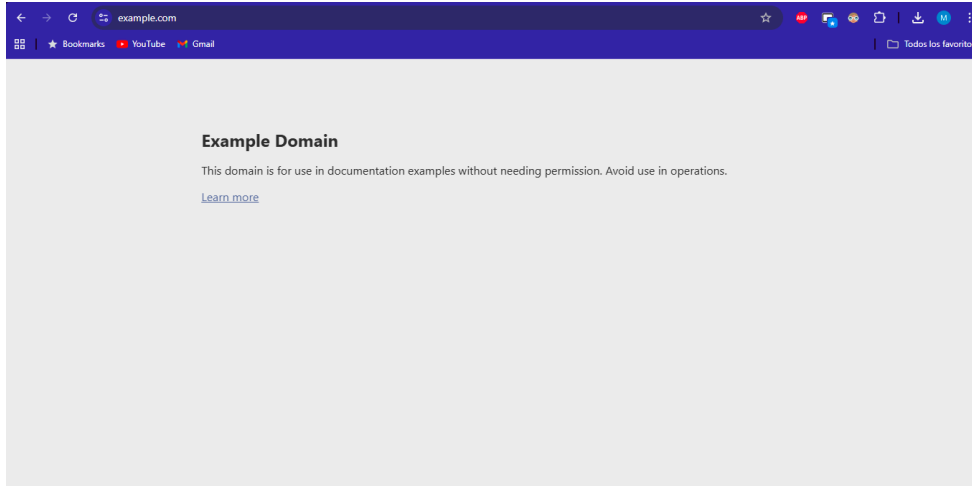
Viernes 25 de Marzo del 2026

Saltillo, Coahuila

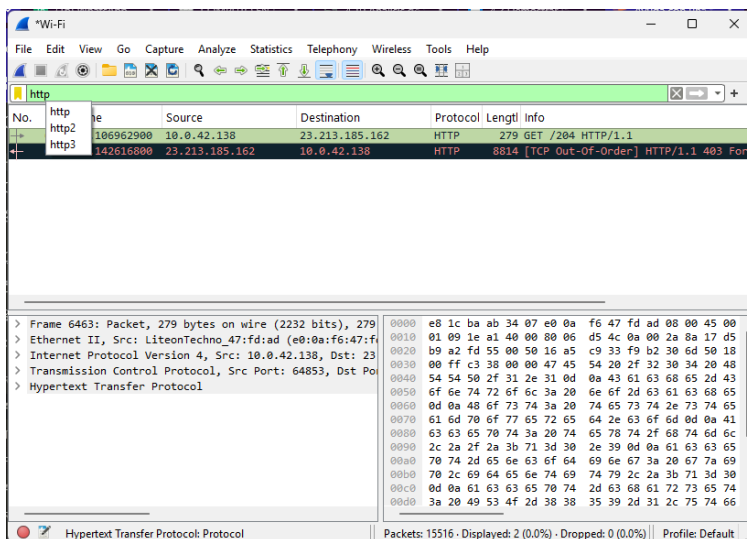
Desarrollo de la práctica

- Abrir Wireshark
- Seleccionar interfaz activa: WiFi
- Clic en Start Capturing • Abrir navegador y entrar a: <http://example.com> (HTTP sin cifrado) y <https://google.com> (HTTPS)
- Clic en botón rojo (Stop)

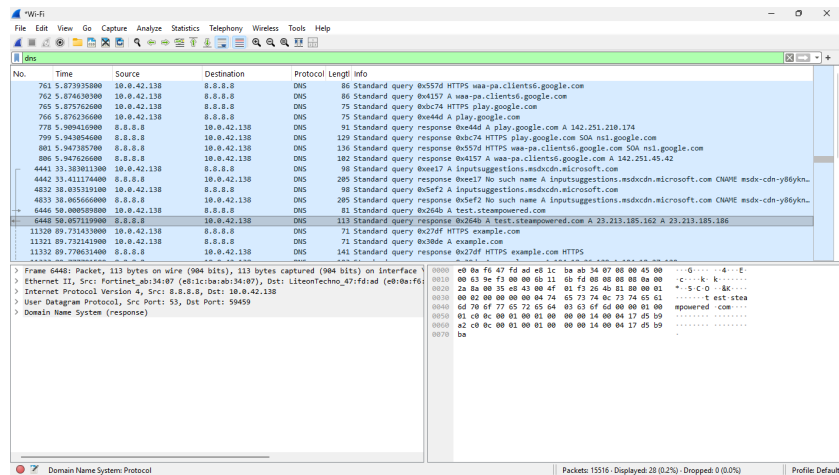




Filtrar tráfico HTTP

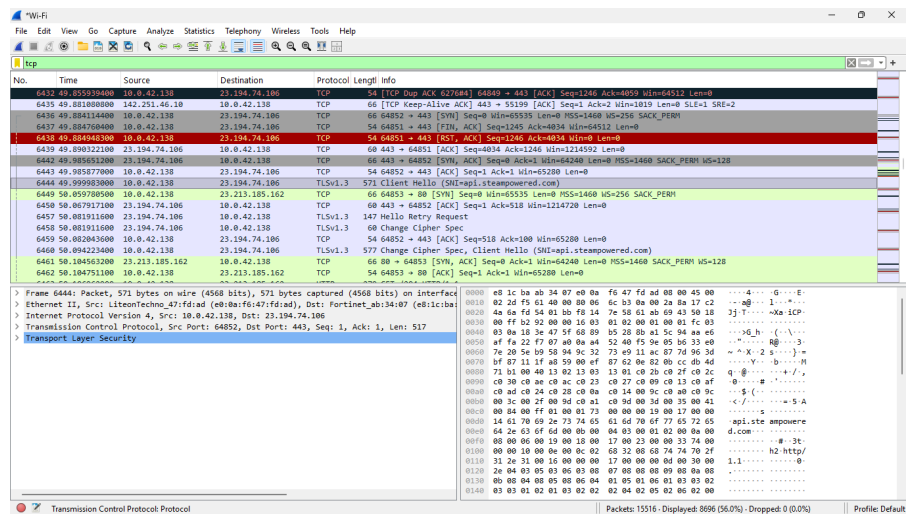


Filtrar DNS



No.	Time	Source	Destination	Protocol	Length	Info
761	5.873935000	10.0.42.138	8.8.8.8	DNS	86	Standard query 0x557d HTTPS waa-pa.clients6.google.com
762	5.874630000	10.0.42.138	8.8.8.8	DNS	86	Standard query 0x4157 A waa-pa.clients6.google.com
765	5.875762000	10.0.42.138	8.8.8.8	DNS	75	Standard query 0x0c74 HTTPS play.google.com
766	5.876216000	10.0.42.138	8.8.8.8	DNS	75	Standard query 0x0e4d A play.google.com
778	5.900410000	10.0.42.138	8.8.8.8	DNS	91	Standard query response 0x0e4d A play.google.com A 142.251.210.174
799	5.903654000	10.0.42.138	8.8.8.8	DNS	129	Standard query response 0x0c74 HTTPS play.google.com SOA ns1.google.com
801	5.947357000	10.0.42.138	8.8.8.8	DNS	136	Standard query response 0x557d HTTPS waa-pa.clients6.google.com SOA ns1.google.com
806	5.947626000	10.0.42.138	8.8.8.8	DNS	182	Standard query response 0x4157 A waa-pa.clients6.google.com A 142.251.45.42
4441	33.303611000	10.0.42.138	8.8.8.8	DNS	98	Standard query 0x0e17 A inputsuggestions.msdcn.microsoft.com
4442	33.411174000	10.0.42.138	8.8.8.8	DNS	285	Standard query response 0x0e17 No such name A inputsuggestions.msdcn.microsoft.com CNAME msdn-cdn-yb8ykn
4832	38.035319000	10.0.42.138	8.8.8.8	DNS	98	Standard query 0x5ef2 A inputsuggestions.msdcn.microsoft.com
4833	38.065660000	10.0.42.138	8.8.8.8	DNS	285	Standard query response 0x5ef2 No such name A inputsuggestions.msdcn.microsoft.com CNAME msdn-cdn-yb8ykn
6446	50.008598000	10.0.42.138	8.8.8.8	DNS	81	Standard query 0x264b A test.steampowered.com
6448	50.057119000	10.0.42.138	8.8.8.8	DNS	113	Standard query response 0x264b A test.steampowered.com A 23.213.185.162 A 23.213.185.186
11320	89.114143000	10.0.42.138	8.8.8.8	DNS	71	Standard query 0x270f HTTPS example.com
11321	89.732419000	10.0.42.138	8.8.8.8	DNS	71	Standard query 0x30de A example.com
11332	89.786234000	10.0.42.138	8.8.8.8	DNS	141	Standard query response 0x270f HTTPS example.com HTTPS

Filtrar tráfico TCP



No.	Time	Source	Destination	Protocol	Length	Info
6432	49.555334000	10.0.42.138	23.194.74.106	TCP	54	TCP Dup ACK 6770484 Seq=6443 [ACK] Seq=6443 Ack=64512 Len=0
6435	49.621800000	142.251.46.10	10.0.42.138	TCP	60	[TCP keep-alive ACK] 443 → 55159 [ACK] Seq=1 Ack=3 Win=1019 Len=0 Sli=1 SRe=2
6436	49.684114000	10.0.42.138	23.194.74.106	TCP	68	64852 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
6437	49.684708000	10.0.42.138	23.194.74.106	TCP	54	64851 → 443 [FIN, ACK] Seq=1245 Ack=4834 Win=64512 Len=0
6438	49.684708000	10.0.42.138	23.194.74.106	TCP	54	64852 → 443 [ACK] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
6439	49.690221000	10.0.42.138	23.194.74.106	TCP	68	443 → 64851 [ACK] Seq=4834 Ack=1246 Win=1214592 Len=0
6442	49.985651200	23.194.74.106	10.0.42.138	TCP	66	443 → 64852 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
6443	49.985677000	10.0.42.138	23.194.74.106	TCP	54	64852 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
6444	49.995033000	10.0.42.138	23.194.74.106	TLSv1.3	571	Client Hello [SH:apli-steampowered.com]
6449	50.057708000	10.0.42.138	23.213.185.162	TCP	68	64853 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
6450	50.067917000	10.0.42.138	23.194.74.106	TCP	68	443 → 64852 [ACK] Seq=1 Ack=1 Win=1214720 Len=0
6457	50.081911000	23.194.74.106	10.0.42.138	TLSv1.3	147	Hello Retry Request
6458	50.081911000	23.194.74.106	10.0.42.138	TLSv1.3	68	Change Cipher Spec
6459	50.082043000	10.0.42.138	23.194.74.106	TCP	54	64852 → 443 [ACK] Seq=518 Ack=100 Win=65280 Len=0
6460	50.094223000	10.0.42.138	23.194.74.106	TLSv1.3	577	Change Cipher Spec, Client Hello [SH:apli-steampowered.com]
6461	50.104563000	23.213.185.162	10.0.42.138	TCP	68	80 → 64853 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
6462	50.104751100	10.0.42.138	23.213.185.162	TCP	54	64853 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0

Comparar HTTP vs HTTPS

¿Qué información se ve en HTTP?

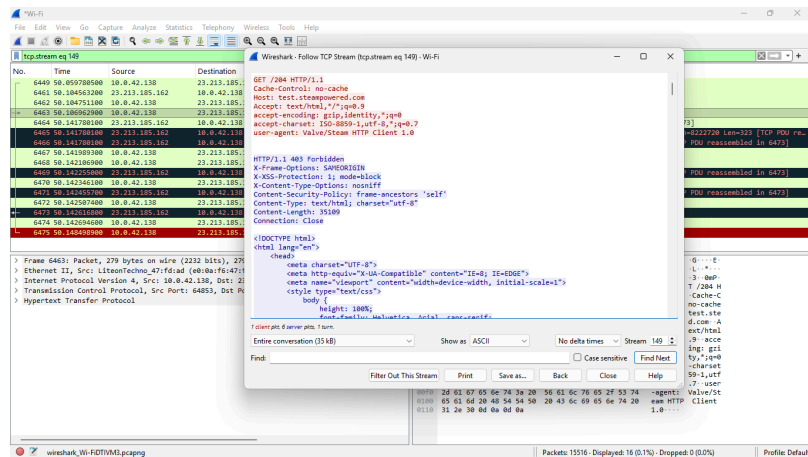
Se puede ver las URL completas, los métodos de solicitud, encabezados de HTTP, cookies de sesión, credenciales de usuario e incluso contenido completo de la página

¿Qué NO se ve en HTTPS?

Todo está cifrado así que no hay mucho que ver más que la ip del servidor, el dominio general, la cantidad de datos transferidos, pero no su contenido

Ver textos en claro

- Seleccionar un paquete HTTP:
- Click derecho → Follow → TCP Stream



¿Qué riesgo existe?

Cualquier persona conectada a la misma red WiFi puede usar WireShark para interceptar y leer toda la comunicación HTTP. Esto se conoce como ataque Man-in-the-middle(MitM).En redes públicas esto especialmente peligroso ya que no se requieren permisos especiales para capturar el tráfico.